

1 Solution — GIAM 4.3

Problem 3

1.1 Problem

Do element-chasing proofs — show that an element is in the left-hand side if and only if it is in the right-hand side — to prove each of the following set equalities:

- **a.** $\overline{A \cap B} = \overline{A} \cup \overline{B}$
- **b.** $A \cup B = A \cup (\overline{A} \cap B)$
- **c.** $A \triangle B = (A \cup B) \setminus (A \cap B)$
- **d.** $(A \cup B) \setminus C = (A \setminus C) \cup (B \setminus C)$

1.2 The Method

$$\boxed{x \in \text{LHS} \iff \dots \iff x \in \text{RHS}}$$

Each proof is a chain of **equivalences**. Because every link runs in both directions, a single chain establishes $\text{LHS} \subseteq \text{RHS}$ and $\text{RHS} \subseteq \text{LHS}$ simultaneously, which by extensionality gives equality. The steps themselves are the definitions of $\cap, \cup, \setminus, \overline{}$ together with tautologies of propositional logic.

All four identities were additionally checked over every subset of a five-element universe — 32×32 pairs for **a**, **b**, **c** and $32^3 = 32768$ triples for **d** — with no exceptions.

1.3 Proof of a

$$\begin{array}{ll}
 x \in \overline{A \cap B} & \iff \neg(x \in A \cap B) & \text{complement} \\
 & \iff \neg(x \in A \wedge x \in B) & \cap \\
 & \iff \neg(x \in A) \vee \neg(x \in B) & \text{De Morgan} \\
 & \iff x \in \overline{A} \vee x \in \overline{B} & \text{complement} \\
 & \iff x \in \overline{A} \cup \overline{B} & \cup
 \end{array}$$

Every line is an equivalence, so this single chain gives both containments. ■

1.4 Proof of b

$$\begin{aligned}
 x \in A \cup B &\iff x \in A \vee x \in B && \cup \\
 &\iff x \in A \vee (\neg(x \in A) \wedge x \in B) && (*) \\
 &\iff x \in A \vee (x \in \overline{A} \cap B) && \text{complement, } \cap \\
 &\iff x \in A \cup (\overline{A} \cap B) && \cup
 \end{aligned}$$

■ The starred step is the only non-obvious one, and it is a tautology:

$$p \vee (\neg p \wedge q) \equiv (p \vee \neg p) \wedge (p \vee q) \equiv \mathbf{t} \wedge (p \vee q) \equiv p \vee q,$$

by distributivity, the excluded middle, then the identity law.

1.5 Proof of c

Recall that $A \triangle B$ consists of the elements lying in exactly one of A and B .

$$\begin{aligned}
 x \in A \triangle B &\iff (x \in A \wedge x \notin B) \vee (x \in B \wedge x \notin A) && \triangle \\
 &\iff (x \in A \vee x \in B) \wedge \neg(x \in A \wedge x \in B) && (**) \\
 &\iff x \in A \cup B \wedge x \notin A \cap B && \cup, \cap \\
 &\iff x \in (A \cup B) \setminus (A \cap B) && \setminus
 \end{aligned}$$

■ Here $(**)$ is the tautology $p \oplus q \equiv (p \vee q) \wedge \neg(p \wedge q)$: “exactly one” means “at least one, but not both”.

1.6 Proof of d

$$\begin{aligned}
 x \in (A \cup B) \setminus C &\iff x \in A \cup B \wedge x \notin C && \setminus \\
 &\iff (x \in A \vee x \in B) \wedge x \notin C && \cup \\
 &\iff (x \in A \wedge x \notin C) \vee (x \in B \wedge x \notin C) && \text{distributivity} \\
 &\iff x \in A \setminus C \vee x \in B \setminus C && \setminus \\
 &\iff x \in (A \setminus C) \cup (B \setminus C) && \cup
 \end{aligned}$$

Five reversible steps, so the two sides have exactly the same elements. ■

1.7 The Picture

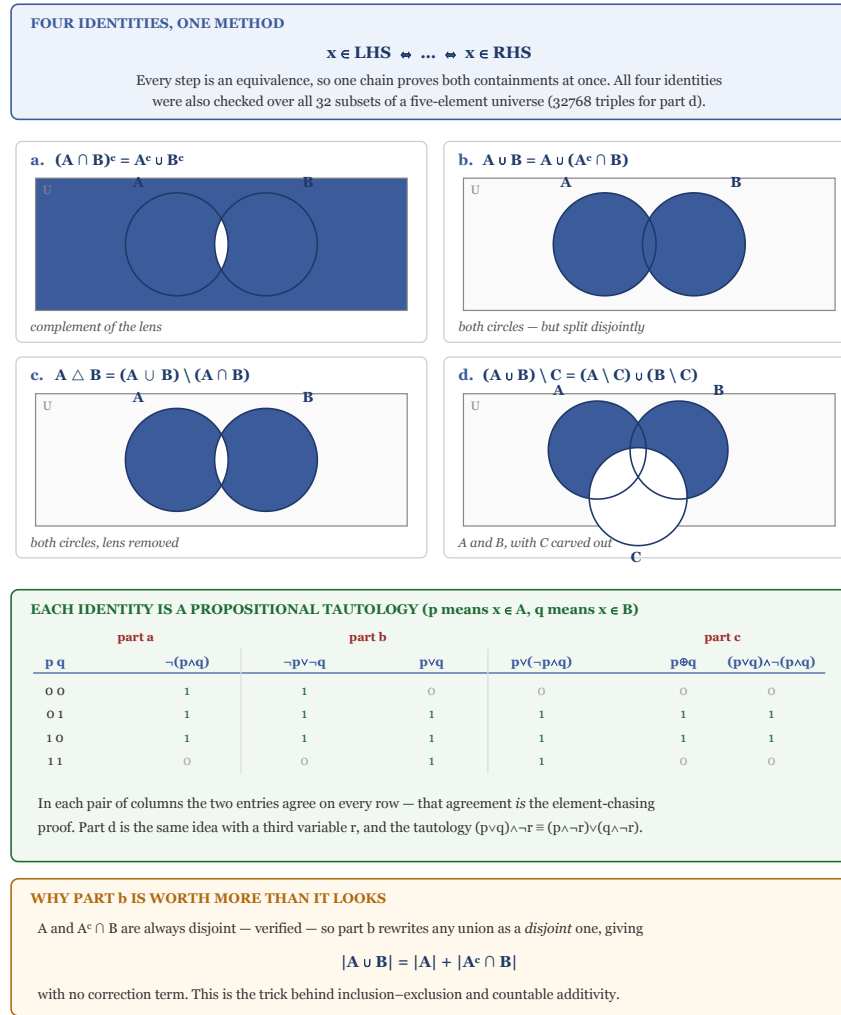


Figure 1.1: A diagram in four parts. The first says four identities, one method: x is in the left-hand side if and only if dot dot dot if and only if x is in the right-hand side. Every step is an equivalence, so one chain proves both containments at once, and all four identities were also checked over all 32 subsets of a five-element universe, with 32768 triples for part d. The second part shows four Venn diagrams. Part a, the complement of A intersect B equals the complement of A union the complement of B , shades the whole universe rectangle except the lens where the two circles overlap. Part b, A union B equals A union the complement of A intersect B , shades both circles entirely, noted as both circles but split disjointly. Part c, the symmetric difference of A and B equals A union B minus A intersect B , shades both circles with the lens removed. Part d, A union B minus C equals A minus C union B minus C , draws three circles and shades A and B with the circle C carved out in white. A green band shows the truth tables that make each identity a propositional tautology, where p means x is in A and q means x is

in B . For part a the columns $\text{not } p$ and q , and $\text{not } p$ or $\text{not } q$, agree on all four rows. For part b the columns p or q , and p or the conjunction of $\text{not } p$ with q , agree on all four rows. For part c the columns p exclusive-or q , and p or q and $\text{not } p$ and q , agree on all four rows. The band notes that in each pair of columns the two entries agree on every row, and that this agreement is the element-chasing proof, with part d the same idea using a third variable r and the tautology that p or q and $\text{not } r$ is equivalent to p and $\text{not } r$, or q and $\text{not } r$. An orange band explains why part b is worth more than it looks: A and the complement of A intersect B are always disjoint, verified, so part b rewrites any union as a disjoint one, giving the size of A union B equals the size of A plus the size of the complement of A intersect B with no correction term, which is the trick behind inclusion and exclusion and countable additivity.

1.8 Remark — These Are Logical Tautologies Wearing Set Clothing

Every step above that was not a definition was a law of propositional logic. Writing p for “ $x \in A$ ”, q for “ $x \in B$ ” and r for “ $x \in C$ ”, the four identities are literally

$$\begin{array}{l|l} \mathbf{a} & \neg(p \wedge q) \equiv \neg p \vee \neg q \\ \mathbf{b} & p \vee q \equiv p \vee (\neg p \wedge q) \\ \mathbf{c} & p \oplus q \equiv (p \vee q) \wedge \neg(p \wedge q) \\ \mathbf{d} & (p \vee q) \wedge \neg r \equiv (p \wedge \neg r) \vee (q \wedge \neg r) \end{array}$$

all four confirmed by truth table. This is the dictionary from Chapter 2 read in the other direction: there, set operations illustrated the connectives; here, the connectives *prove* facts about sets. The reason the translation is exact is that membership is two-valued — x is in a set or it is not, with no third option — so a set is nothing more than a truth-valued function on the universe.

A practical consequence: any propositional tautology in n variables yields a set identity in n sets, and vice versa. There is no need to memorise set identities separately from logical ones — and a proposed identity can be tested by truth table before any attempt is made to prove it.

The dictionary also comes with a free symmetry. Interchanging $\wedge$ with $\vee$ and $\mathbf{t}$ with $\mathbf{c}$ turns any tautology into another tautology, so interchanging $\cap$ with $\cup$ and $\emptyset$ with U turns any set identity into another — its **dual**. Part **a** dualises to $\overline{A \cup B} = \overline{A} \cap \overline{B}$ and

al chain with the symbols swapped. Half of any table of set identities is therefore redundant.

1.9 Remark — Why the Chain Must Run Both Ways

The instruction “show that an element is in the left-hand side if and only if it is in the right-hand side” is doing real work. A chain of one-way implications

$$x \in \text{LHS} \implies \cdots \implies x \in \text{RHS}$$

proves only $\text{LHS} \subseteq \text{RHS}$, which is strictly weaker. For instance $x \in A \implies x \in A \cup B$ always holds, but $A = A \cup B$ holds only when $B \subseteq A$ — verified for all 32×32 pairs. So the one-way chain would “prove” a false statement if its direction were forgotten.

Every step in the four proofs above is reversible because each is either a definition — which is an equivalence by fiat — or a logical equivalence $\equiv$. When a step is not obviously reversible, the honest thing is to split into two separate containment proofs, which is the pattern used in 4.2 Problem 6.

1.10 Remark — Part b Is the Disjointification Trick

Parts **a**, **c** and **d** are standard rearrangements, but **b** is the useful one in practice. Its right-hand side is a union of two **disjoint** pieces:

$$A \cap (\overline{A} \cap B) = \emptyset,$$

verified for all pairs. Since sizes add over disjoint unions,

$$|A \cup B| = |A| + |\overline{A} \cap B|,$$

also verified — with no correction term, unlike inclusion–exclusion. Part **b** is how one *derives* inclusion–exclusion rather than assuming it: $\overline{A} \cap B = B \setminus (A \cap B)$, so the second term is $|B| - |A \cap B|$.

The same manoeuvre, applied to an infinite sequence, turns $\bigcup_n E_n$ into a disjoint union by replacing E_n with $E_n \setminus (E_1 \cup \cdots \cup E_{n-1})$. That is exactly the step that makes

countable additivity usable in measure theory and probability, and it is nothing more than part **b** applied repeatedly.

1.11 Remark — Part **d** Says Removing C Is Structure-Preserving

Part **d** states that the operation $X \mapsto X \setminus C$ commutes with union. In fact $X \setminus C = X \cap \overline{C}$, and intersecting with a fixed set preserves everything:

$$\begin{aligned}(X \cup Y) \setminus C &= (X \setminus C) \cup (Y \setminus C) \\ (X \cap Y) \setminus C &= (X \setminus C) \cap (Y \setminus C) \\ (X \triangle Y) \setminus C &= (X \setminus C) \triangle (Y \setminus C)\end{aligned}$$

all three verified over all 32^3 triples. In the language of 4.3 Problem 1, the third line says $X \mapsto X \setminus C$ is a ring homomorphism from $\mathcal{P}(U)$ onto $\mathcal{P}(\overline{C})$ for the Boolean ring structure $(\triangle, \cap)$.

So part **d** is not an isolated curiosity: it is one instance of the general fact that restricting attention to a subuniverse — ignoring everything in C — cannot change the answer to any question built from $\cup$, $\cap$ and $\triangle$. Note that complement is the exception, since $\overline{X} \setminus C$ and $\overline{X \setminus C}$ differ; complementation depends on which universe one is working in.